REHABGRIP

MULTIMODAL HAND REHABILITATION MEASUREMENT SYSTEM — MODEL RG-2026

1. SENSING CHANNELS

PARAMETER	SPECIFICATION
Flex sensors	5 channels, 2.2-inch (resistive)
ADC resolution	12-bit (ESP32-S3)
Sampling rate	50 Hz per channel
Angle range	0° – 90° (calibrated)
Angle accuracy	±2° (post-calibration)
EMG channels	1 surface EMG (Muscle BioAmp)
EMG bandwidth	72 – 720 Hz
EMG gain	×2420 (fixed)
IMU axes	6 (3 accelerometer, 3 gyroscope)
Accel range	±2g / ±4g / ±8g
Gyro range	±250°/s – ±2000°/s
IMU interface	I²C, 400 kHz

2. SIGNAL PROCESSING

PARAMETER	SPECIFICATION
Microcontroller	ESP32-S3, dual core 240 MHz
Flash memory	8 MB (on-board)
RAM	512 KB SRAM
Wireless	BLE 5.0, WiFi 2.4 GHz
Data format	JSON over WebSocket (secure)
Latency	< 80 ms end-to-end
Display	SH1106 OLED, 132 × 64 pixels
Display interface	I²C (0x3C)
On-device processing	Filtering, basic feature extraction
Cloud integration	Render (API/WebSocket)

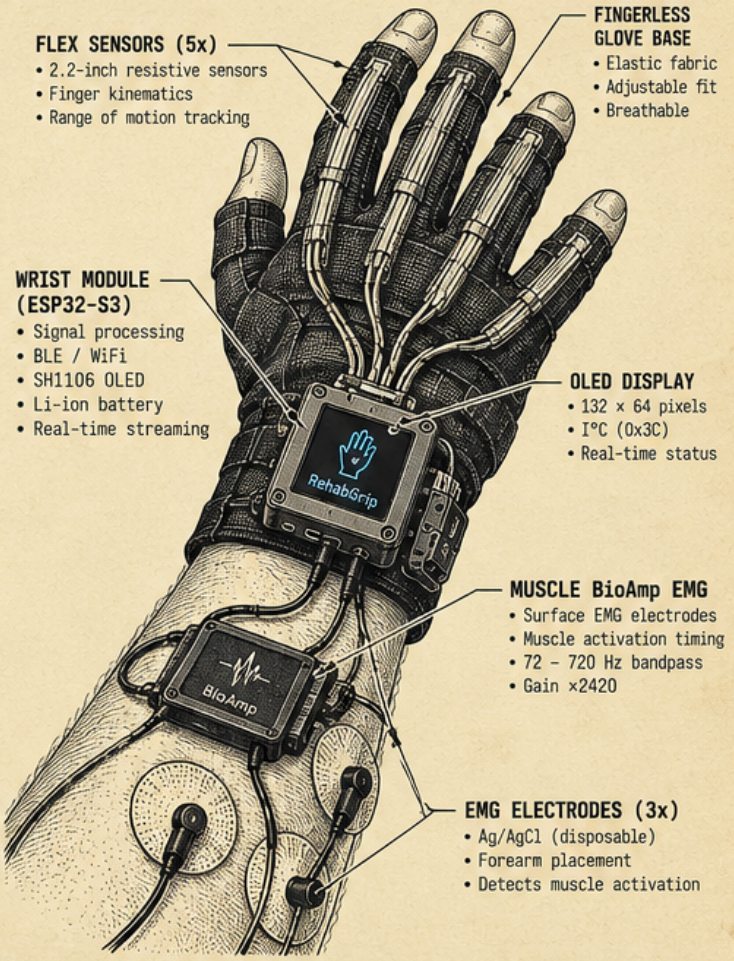
3. POWER

PARAMETER	SPECIFICATION
Battery	18650 Li-ion, 3.7 V
Capacity	2600 mAh (typ.)
Operating time	~ 4 hours (continuous use)
Charging	USB-C via TP4056
Charge time	~ 2 hours (0–100%)
Voltage regulation	3.3 V (AMS1117-3.3)
Feedback	Haptic (coin vibration motor) + Visual (5 mm LED)
Power indicator	On-board OLED / LED

4. MECHANICAL

PARAMETER	SPECIFICATION
Enclosure (wrist)	85 × 55 × 28 mm (L × W × H)
EMG housing	55 × 40 × 25 mm (forearm)
Material	3D printed PLA (matte black)
Glove base	Fingerless, elastic fabric
Total weight (glove + electronics)	~ 180 g
Wrist strap	Velcro, adjustable
EMG electrodes	Disposable Ag/AgCl (3 per session)
Cable routing	Strain relief channels

5. SYSTEM OVERVIEW



6. SOFTWARE & CONNECTIVITY

- Real-time 3D hand visualization (Three.js)
- Per-finger ROM, velocity, consistency tracking
- EMG signal envelope + flex synchronization
- Session history, trend charts, PDF reports
- AI movement assessment (10 condition profiles)
- Patient management (Indian clinical workflow)
- Render cloud relay — WebSocket streaming
- Vercel-hosted clinical dashboard

7. CLINICAL APPLICATIONS

- Post-stroke upper limb rehabilitation
- Parkinson's disease motor assessment
- Hand osteoarthritis ROM monitoring
- Carpal tunnel syndrome rehabilitation
- Traumatic brain injury motor recovery
- Research: multimodal hand biomechanics

8. PERFORMANCE INDICES (COMPARISON)

METRIC	REHABGRIP	MANUAL ASSESSMENT
ROM accuracy	±2°	±10 – 15°
Sampling frequency	50 Hz	1 × / session
EMG capture	Yes	No
Wrist compensation	Detected	Undetected
Session data	100% (time-series)	< 5% recorded
Home use	Yes	No